NURobotics Robotics Software Course

Week 4: Probability & Particle Filters

Set & Function Notation

Implication

$(a \Rightarrow b)$

If *a* is true then *b* must be true

“*a* implies *b*”

Biconditional

($a \Leftrightarrow b$)

If *a* is true then *b* must be true, and if *b* is true, then *a* must be true

“*a* if and only if *b*”

Set

A collection of objects. Each object in the set is called an *element*

$A = \{1, 4, 7, 9, 12\}$

Cardinality

The number of elements in a set

$|A|$

$A = \{1, 4, 7, 9, 12\}$

$|A| = 5$

Membership

$\in$: is an element of

$\notin$: is **not** an element of

$A = \{1, 4, 7, 9, 12\}$

$1 \in A$

$2 \notin A$

Special Sets

$\varnothing$: The null or empty set

$\mathbb{Z}$: The set of all integers

$\mathbb{Z}^+$: The set of all positive integers

$\mathbb{R}$: The set of all real numbers

For All

This is true for all elements

$\forall$

$\forall x \in \mathbb{Z}^+ \quad x > 0$

There Exists

There is a single element for which this is true

$\exists$

$\exists x \in \mathbb{Z}$ s.t. x is 0

Complement

The set of elements not in A.

A^C

$\forall a \in A : a \notin A^C$

$\Omega = \{1, 2, 3, 4, 5\}$ $A = \{1, 2, 5\}$ $A^C = \{3, 4\}$

Union

$\cup$

$a \in A, b \in B$

$D = A \cup B \Rightarrow \forall d \in D : d \in A \text{ or } d \in B$

$A = \{1, 2, 3\} \quad B = \{2, 3, 4\}$

$A \cup B = \{1, 2, 3, 4\}$

Intersection

$\cap$

$a \in A, b \in B$

$D = A \cap B \Rightarrow \forall d \in D : d \in A \text{ and } d \in B$

$A = \{1, 2, 3\} \quad B = \{2, 3, 4\}$

$A \cap B = \{2, 3\}$

Subset

$\subset$

$a \in A, b \in B$

$A \subset B \Rightarrow \forall a \in A : a \in B$

$A = \{2, 3\} \quad B = \{2, 3, 4\}$

Sets of Vectors/Matrices

$x \in \mathbb{R}^n$

$X \in \mathbb{R}^{n \times n}$

$A \in \mathbb{R}^{2 \times 3}$ $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$

$x^T \in \mathbb{R}^3$ $A = [x_1 \; x_2 \; x_3]$

Functions

Takes in a variable and outputs something

$f(a) : A \rightarrow B$, “f is a mapping from A to B”

$x \in \mathbb{R}^n$

$f(x) : \mathbb{R}^n \rightarrow \mathbb{R}^m$

$y = f(x) \Rightarrow y \in \mathbb{R}^m$

Function With Multiple Inputs

$h(x, z) : \mathbb{R}^n \times \mathbb{R}^k \rightarrow \mathbb{R}^m$

$x \in \mathbb{R}^n$

$z \in \mathbb{R}^k$

$y = h(x, z) \Rightarrow y \in \mathbb{R}^m$

Defining Sets Notation

$f(x) : \mathbb{R}^n \rightarrow \mathbb{R}$

$X = \{x | x \in \mathbb{R}^n, f(x) > 0\}$

$X = \{x \in \mathbb{R}^n, f(x) > 0\}$

Probability

Stochastic

Stochastic = Random



Vocabulary

Random Variable: A random value we want to estimate (denoted by capital letters). Can be discrete or continuous

Experiment: How we measure the random variable

Event: A possible value of a random variable

Sample Space: (Ω) The set of all possible events

Coin Toss

X: Random variable for the coin toss

$\Omega = \{H, T\}$

$P(X = H) = 0.5, P(X = T) = 0.5$

Probability Properties

- All probabilities are in the range [0, 1]
- The probability of every event in the sample space must sum to one $\Sigma_{A \in \Omega} P(A) = 1$
- Probabilities can be very close to zero and the event can still happen $\Sigma_{i=0}^{\infty} 1/\infty = 1$
- $P(A^C) + P(A) = 1$

Combining Probabilities

A, B are events in the same sample space

$A, B \in \Omega$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Combining Probabilities Example 1

What is the probability of rolling $A = \{1, 2\}$ or rolling $B = \{2, 3\}$

$P(A) = 1/3 \quad P(B) = 1/3$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= 1/3 + 1/3 - 1/6 = 1/2$

Mutually Exclusive

Two events are mutually exclusive if the probability of their intersect is 0

Given A, B

$P(A \cap B) = 0$

Combining Probabilities Example 2

We are rolling a fair 6 sided die. What is the probability of rolling a 2 or a 4? $A = \{2, 4\}$

$P(A) = P(X = 2) + P(X = 4) – P(X = 2 \text{ and } X = 4)$

$= 1/6 + 1/6 – 0 = 1/3$

Using the Complement

We are rolling a fair 6 sided die. What is the probability of **not** rolling a 2 or a 4? $A = \{2, 4\}$

$P(A) = 1/3$

$P(A^C) = 1 - P(A) = 1 - 1/3 = 2/3$

Independent Events

Two events are independent if they have no affect on each other

$P(A \cap B) = P(A) * P(B)$

Examples of independence: Two coin flips

Examples of dependence: Picking colors out of a hat

Fair Double Coin Flip Example

What is the probability of {*H H*} on a double fair coin flip?

$P(H) = 0.5, P(T) = 0.5$

$\Omega = \{H\,H, H\,T, T\,H, T\,T\} \quad |\Omega| = 4$

$P(X = H\,H) = 1/|\Omega| = ¼$

$P\,(X = H\,H) = P(H) * P(H) = 0.5 * 0.5 = 0.25$

Biased Double Coin Flip Example

What is the probability of $\{T\ T\}$ on a double biased coin flip?

$P(H) = 0.8$, $P(T) = 0.2$

$\Omega = \{H\ H, H\ T, T\ H, T\ T\}$

$P(X = T\ T) = P(T) * P(T) = 0.2 * 0.2 = 0.04$

Conditional Probability

What is the probability of A given B

$P(A|B) = P(A \cap B) \,/\, P(B)$

We redefine Ω to be a subset of *B* and only consider events *A* that are elements of *B*

Fair Double Coin Flip Example

What is the probability of {*H H*} on a double fair coin flip?

$P(H) = 0.5$, $P(T) = 0.5$

Define X_1 as the event where the first coin is heads, and X and as the event where both are heads

$P(X_1) = 0.5$

$P(X) = P(X_1) * P(X|X_1) = 0.5 * 0.5 = 0.25$

Conditional Independence

If A and B are independent, $P(A|B) = P(A)$

B provides no information on A

Bayes Rule

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \cup B) = P(B|A)P(A)$$

Probability Density/Mass Function

A function to define a random variable

Mass = Discrete, Density = Continuous

$f(x) = P(X = x)$

$f(x) > 0, x \in \Omega$

$\Sigma_{x \in \Omega} f(x) = 1$

Expectation

Expected value or weighted average of a random variable

$$\mathbb{E}[X] = \sum_{x \in \Omega} u(x) f(x)$$

Where $f(x)$ is the pmf of the random variable and $u(x)$ is the expected return for that value

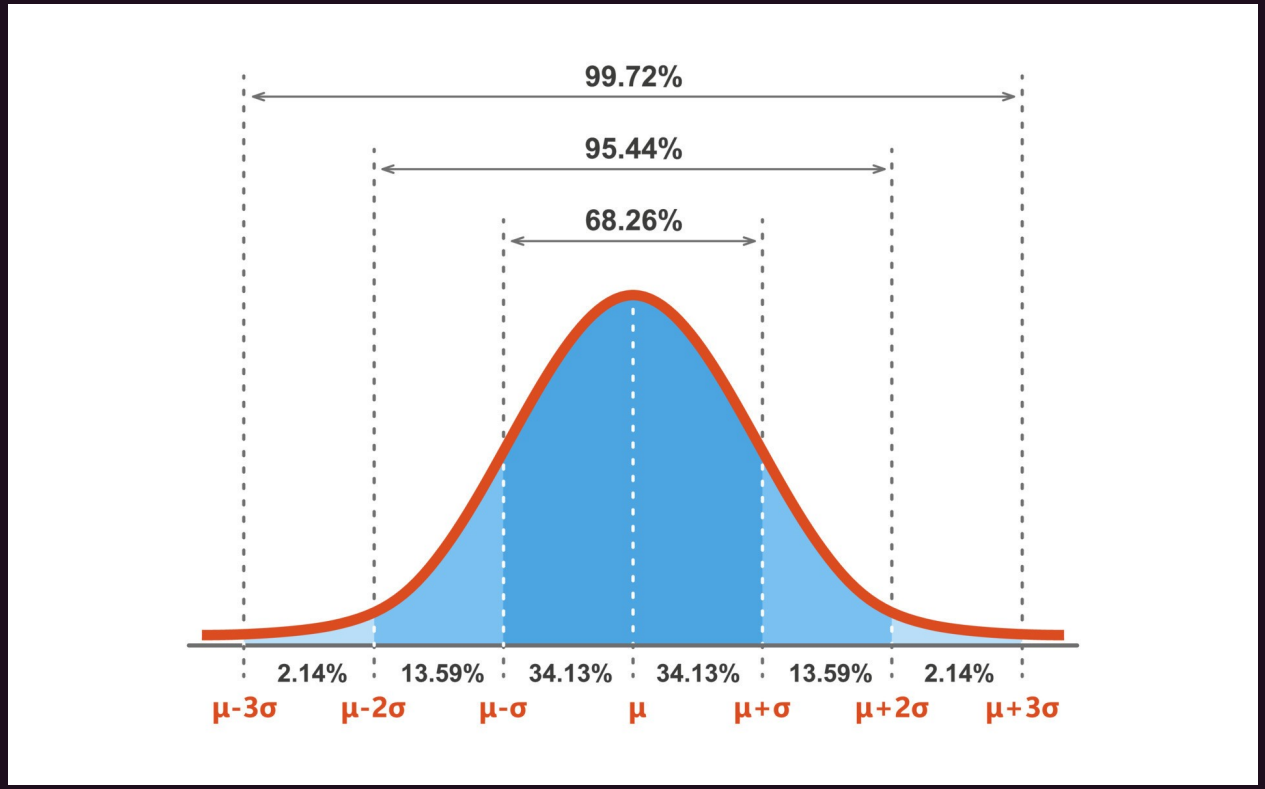
Expected Value Example

What is the expected value if we win a dollar for every heads in a fair coin toss? $X = \{H, T\}$

$f(x) = 0.5 \quad u(H) = 1 \quad u(T) = 0$

$\mathbb{E}[X] = 1 * 0.5 + 0 * 0.5 = 0.5$

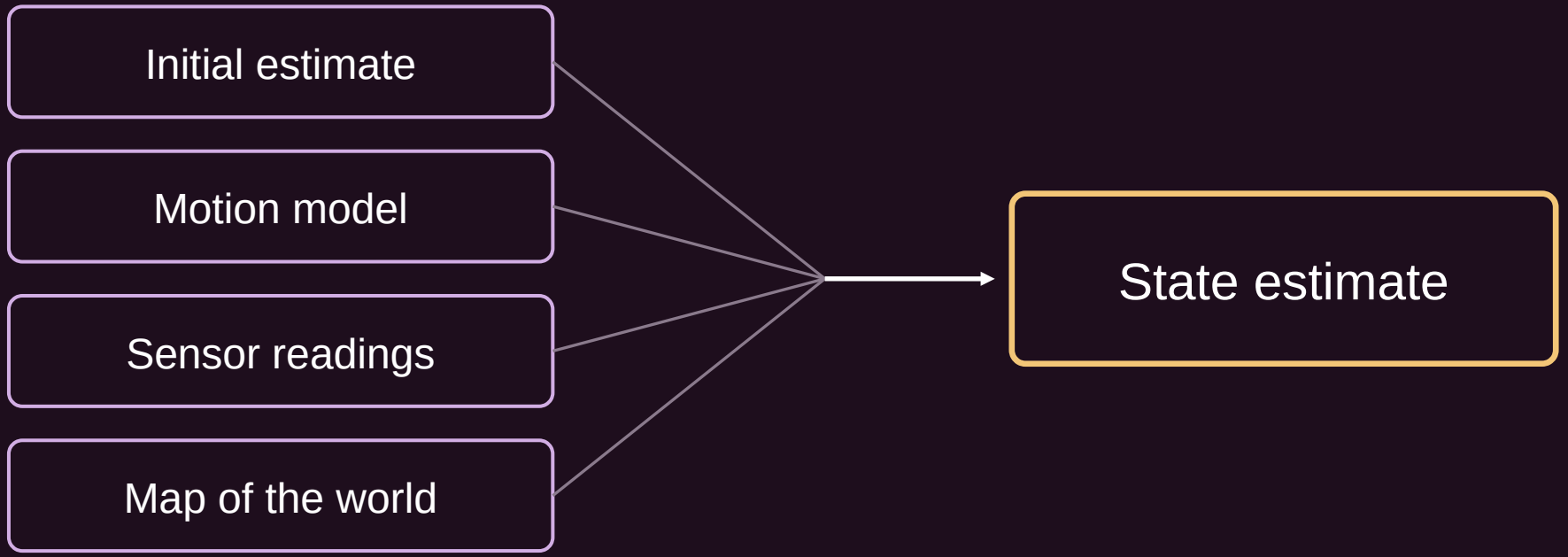
Gaussian (Normal) Distribution



Particle Filter Theory

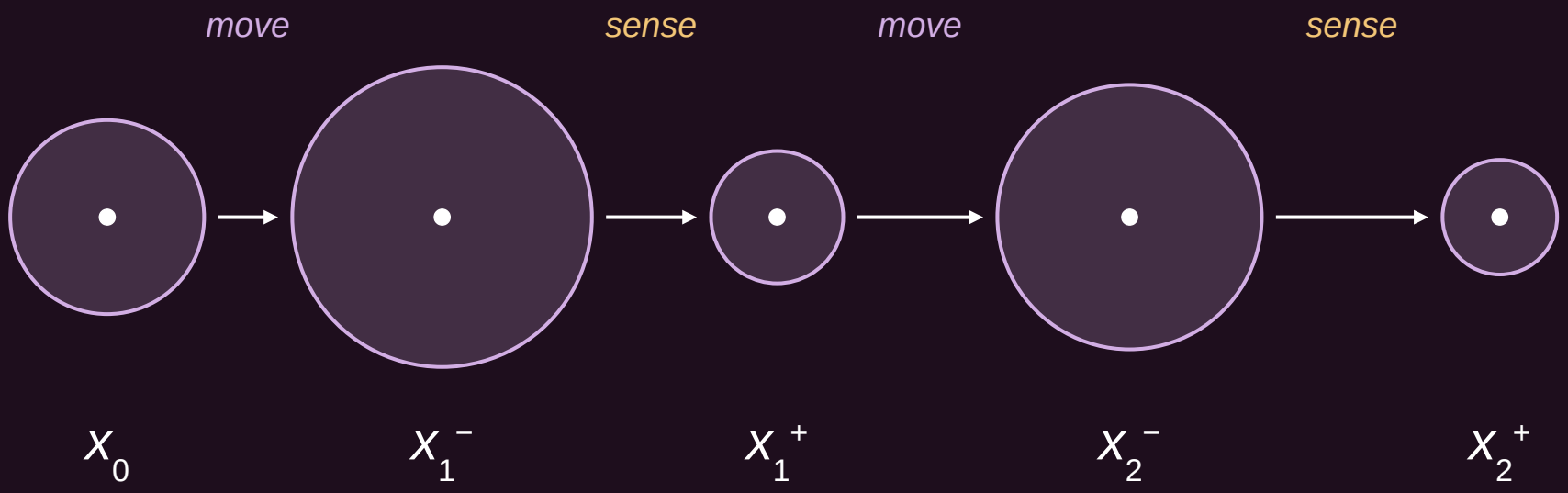
Localization

Where is the robot in the world?



Uncertainty

Moving grows uncertainty; sensing shrinks it

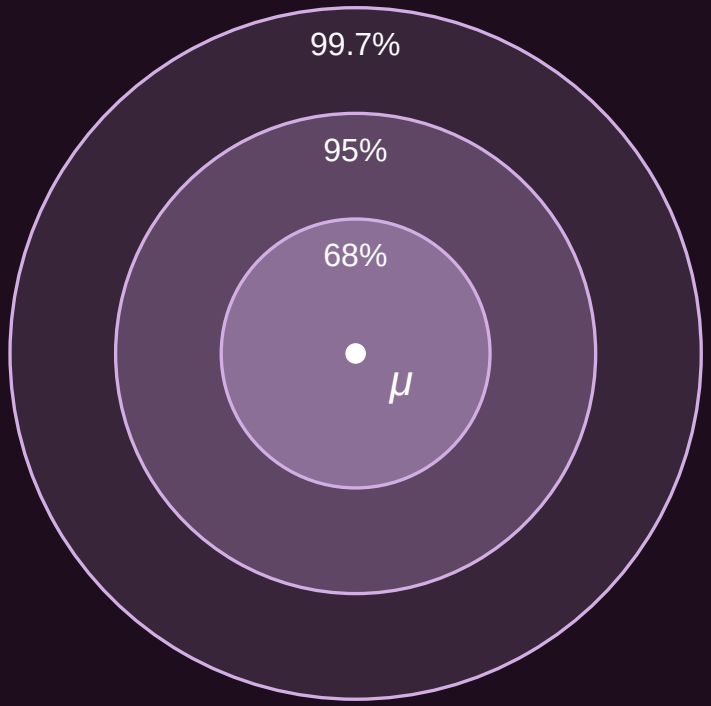


− after moving, before the new sensor reading
+ after applying the sensor reading

Gaussian Assumption

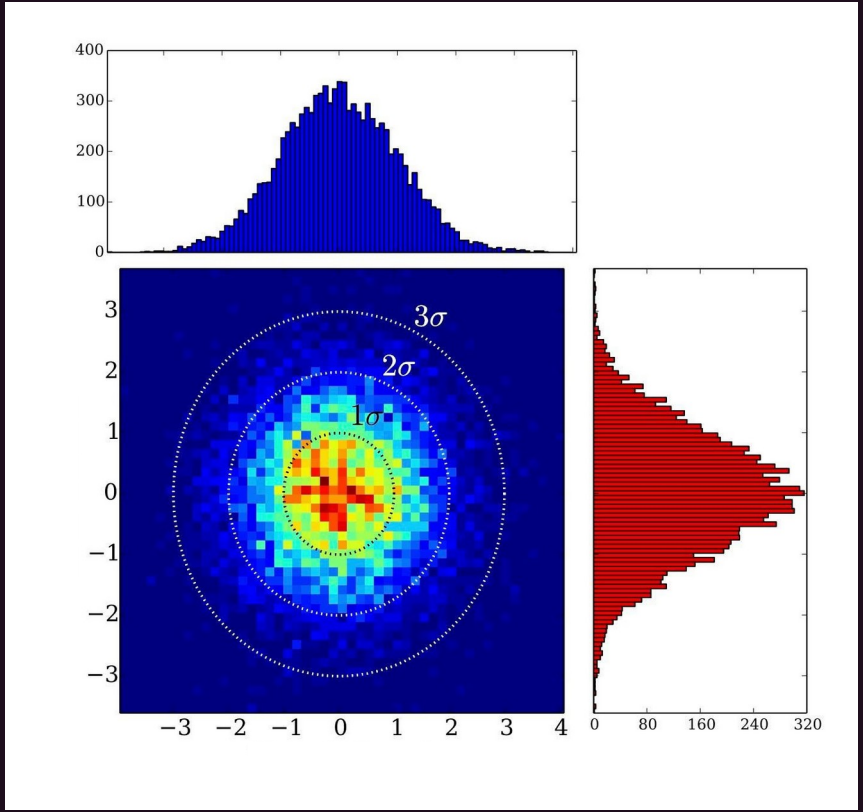
Approximate our uncertainty as a Gaussian

Mean (μ): the center
Variance (σ^2): how far it spreads



Rings at 1σ, 2σ, 3σ

Gaussian Assumption



Goal of State Estimation

Move the mean to the correct location

Keep the variance as low as possible

Notation

$x \in \mathbb{R}^n$: Robot state

$u \in \mathbb{R}^k$: Robot control

$z \in \mathbb{R}^m$: Sensor reading

$\hat{z} \in \mathbb{R}^m$: Estimated sensor reading

$\hat{x} \in \mathbb{R}^n$: Robot state estimate

State Estimation

What is the most probable state given every sensor reading and control so far?

$P(x_t \mid x_{0:t-1}, z_{0:t}, u_{0:t})$

Prior: the information we already know before solving the problem

Sensor Model

Gives the expected sensor reading for a given state

$\hat{z}_t = h(x_t) + \varepsilon$

$P(z_t \mid x_t)$

Where $h(x)$ maps a state to a sensor reading and ε is some stochastic noise

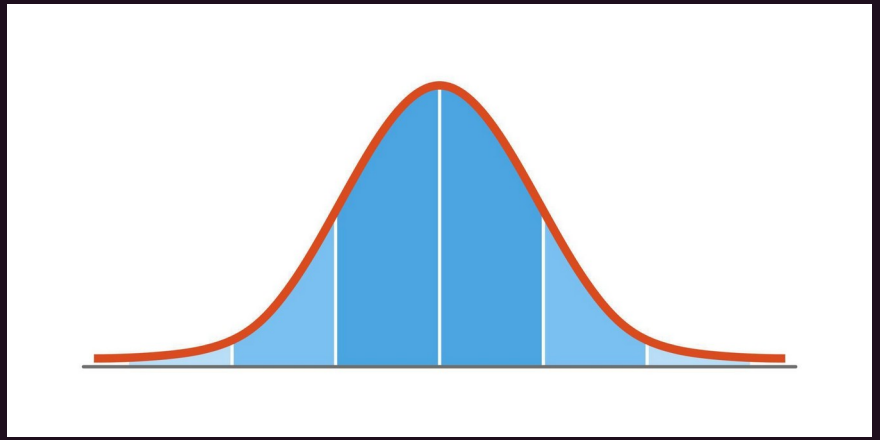
$h(x)$: $\mathbb{R}^n \rightarrow \mathbb{R}^m$

Normal Distribution

Also called the Gaussian distribution

$\varepsilon \sim N(\mu, \sigma^2)$ μ: mean σ^2: variance

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$



Motion Model

Gives the next state given the previous state and the current control (dynamics)

$x_{t+1} = f(x_t, u_t) + w \qquad P(x_{t+1} \mid x_t, u_t)$

Where $f(x, u) : \mathbb{R}^n \times \mathbb{R}^k \rightarrow \mathbb{R}^n$ and w is some stochastic noise

Derivation

Separate out the newest sensor reading

$$P(x_{0:t} \mid z_{0:t}, u_{1:t}) = P(x_{0:t} \mid z_t, z_{0:t-1}, u_{1:t})$$

Derivation: Bayes Rule

$$P(x_{0:t}|z_{0:t}, u_{1:t}) = P(x_{0:t}|z_t, z_{0:t-1}, u_{1:t})$$

Bayes rule with additional conditionals:

$$P(x|y, z) = \frac{P(y|x, z)P(x|z)}{P(y|z)}$$

$$P(x_{0:t}|z_t, z_{1:t-1}, u_{1:t}) = \frac{P(z_t|x_{0:t}, z_{1:t-1}, u_{1:t})P(x_{0:t}|z_{1:t-1}, u_{1:t})}{P(z_t|z_{1:t-1}, u_{1:t})}$$

Derivation: Normalizer

$$P(x_{0:t}|z_{0:t},u_{1:t})=P(x_{0:t}|z_t,z_{0:t-1},u_{1:t})$$

Bayes rule with additional conditionals:

$$P(x|y,z)=\frac{P(y|x,z)P(x|z)}{P(y|z)}$$

$$P(x_{0:t}|z_t,z_{1:t-1},u_{1:t})=\frac{P(z_t|x_{0:t},z_{1:t-1},u_{1:t})P(x_{0:t}|z_{1:t-1},u_{1:t})}{P(z_t|z_{1:t-1},u_{1:t})}$$

The denominator does not depend on the state, so replace it with a constant

$$P(x_{0:t}|z_t,z_{1:t-1},u_{1:t})\propto\eta P(z_t|x_{0:t},z_{1:t-1},u_{1:t})P(x_{0:t}|z_{1:t-1},u_{1:t})$$

Markov Property

$$P(x_{0:t}|z_t, z_{1:t-1}, u_{1:t}) \propto \eta P(z_t|x_{0:t}, z_{1:t-1}, u_{1:t}) P(x_{0:t}|z_{1:t-1}, u_{1:t})$$

The present depends only on the previous step (the state at time *t* only depends on *t*-1

$$P(x_t|x_{0:t-1}, z_{1:t-1}, u_{1:t}) = P(x_t|x_{t-1}, u_t)$$

$$P(z_t|x_{0:t}, z_{1:t-1}, u_{1:t}) = P(z_t|x_t)$$

$$= \eta P(z_t|x_t) P(x_{0:t}|z_{1:t-1}, u_{1:t})$$

$$= \eta P(z_t|x_t) P(x_t|x_{0:t-1}, z_{1:t-1}, u_{1:t}) P(x_{0:t-1}|z_{1:t-1}, u_{1:t})$$

$$= \eta P(z_t|x_t) P(x_t|x_{t-1}, u_t) P(x_{0:t-1}|z_{1:t-1}, u_{1:t})$$

Derivation: Split the Trajectory

Separate the current state from the rest of the history

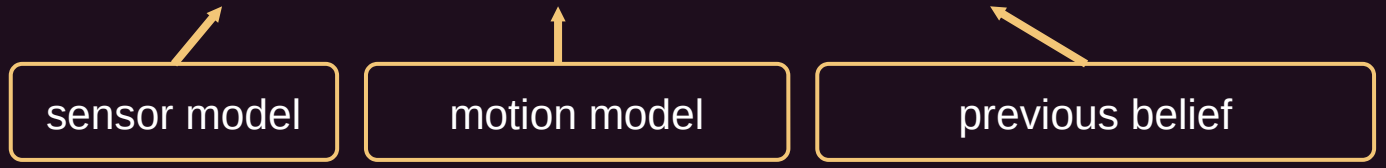
$$
\begin{aligned}
&= \eta\, P(z_t|x_t)\, P(x_{0:t}|z_{1:t-1}, u_{1:t}) \\
&= \eta\, P(z_t|x_t)\, P(x_t|x_{0:t-1}, z_{1:t-1}, u_{1:t})\, P(x_{0:t-1}|z_{1:t-1}, u_{1:t}) \\
&= \eta\, P(z_t|x_t)\, P(x_t|x_{t-1}, u_t)\, P(x_{0:t-1}|z_{1:t-1}, u_{1:t})
\end{aligned}
$$

Result: Recursive Filter

Marginalize over the previous state

$$= \eta P(z_t|x_t) P(x_t|x_{t-1}, u_t) P(x_{0:t-1}|z_{1:t-1}, u_{1:t})$$

$$= \eta \int P(z_t|x_t) P(x_t|x_{t-1}, u_t) P(x_{0:t-1}|z_{1:t-1}, u_{1:t}) dx_{t-1}$$



No closed-form solution in general

Approximate It

Approximate the distribution with many discrete guesses (particles!)

High Level Idea

Given a bunch of guesses of possible states

1. Move the guesses (particles) using the motion model

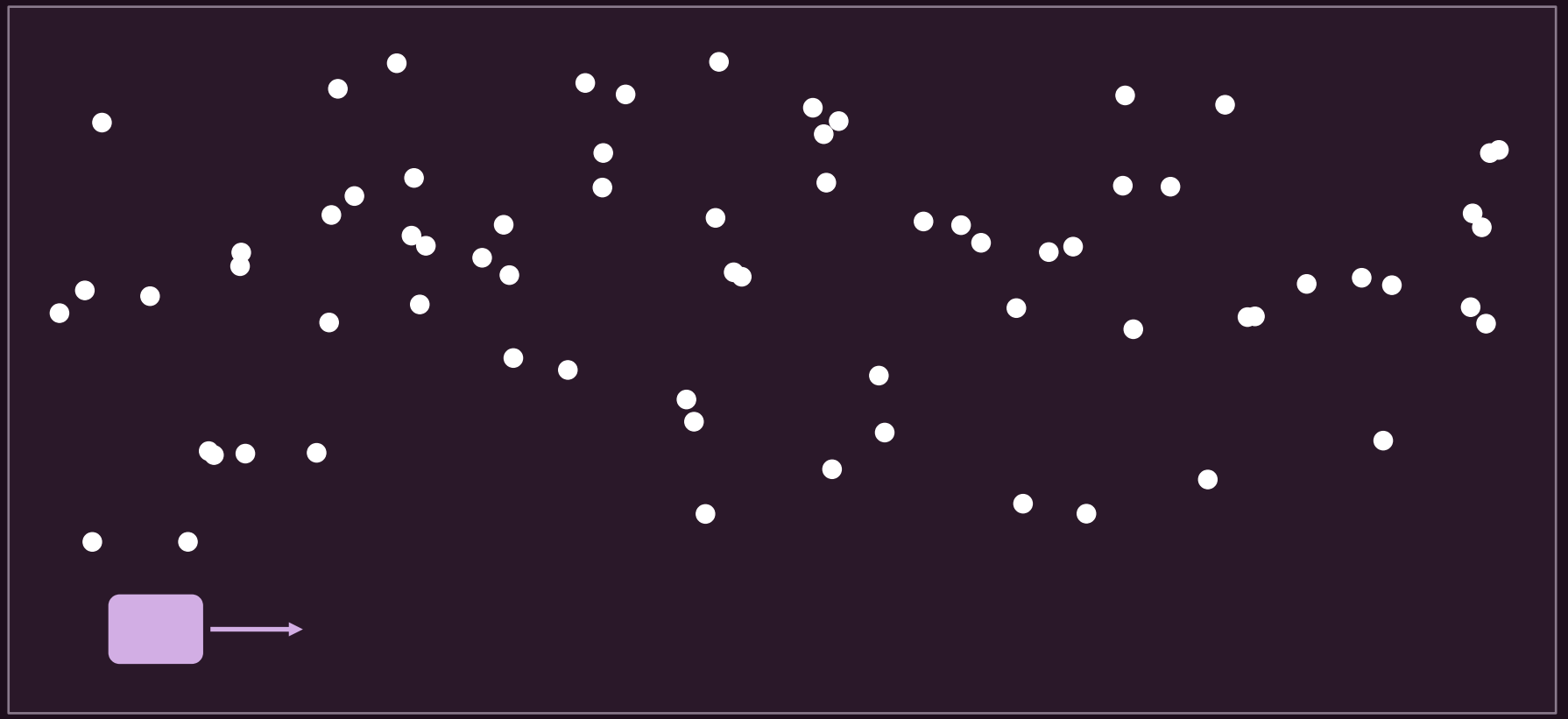
   $P(x_t \mid x_{t-1}, u_t)$

2. Weight the guesses (particles) using the sensor model

   $P(x_t \mid z_t)$

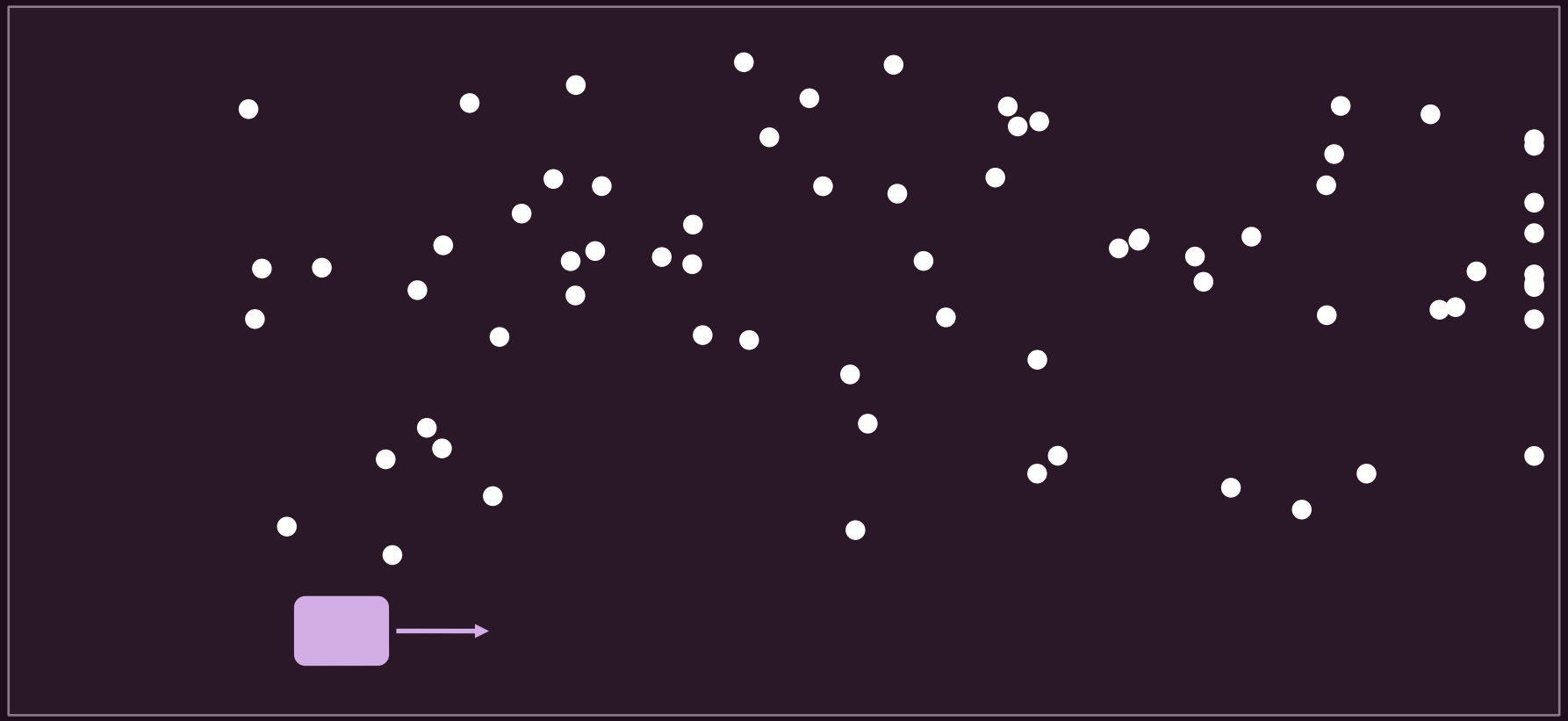
3. Resample toward the more likely particles

Particles: Initial Guesses



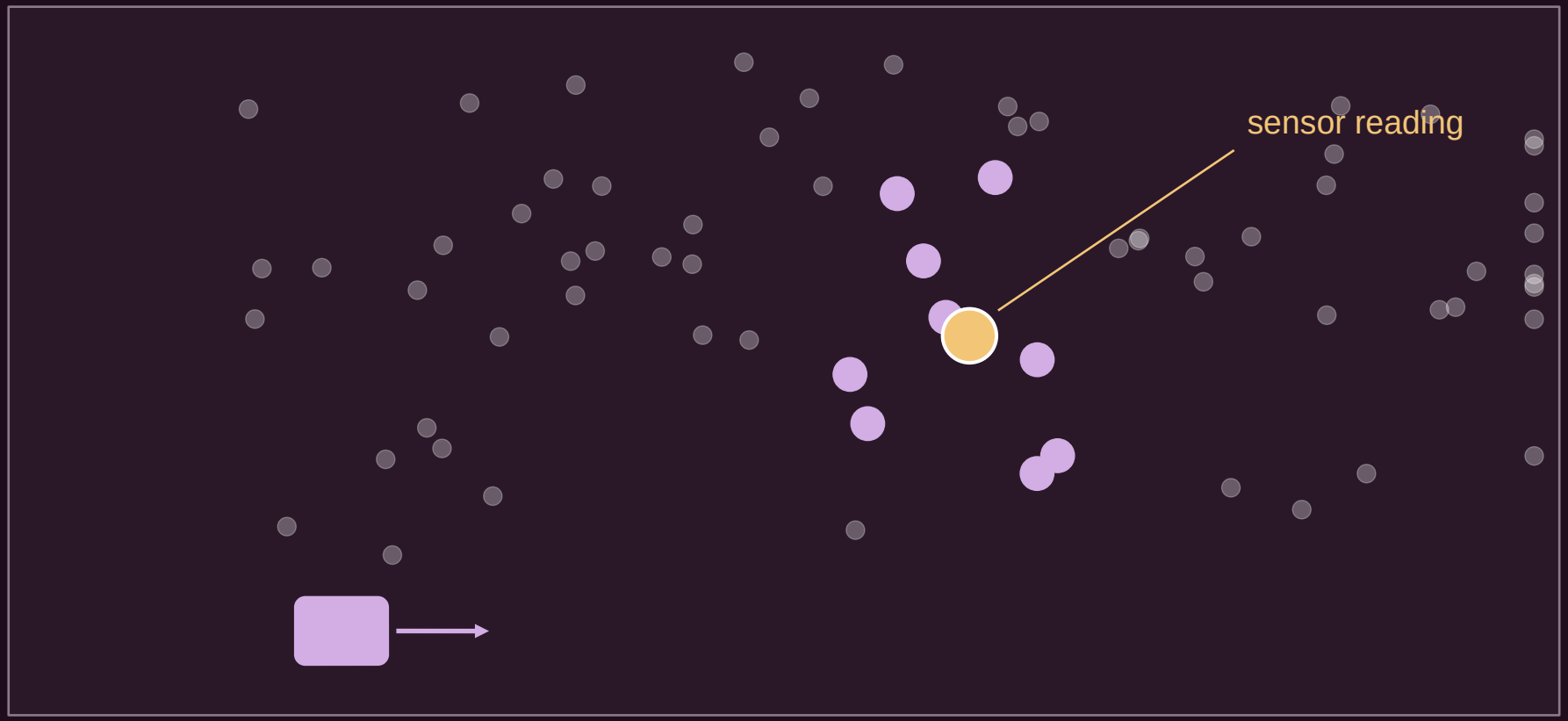
Each dot is a possible robot state

Particles: Move



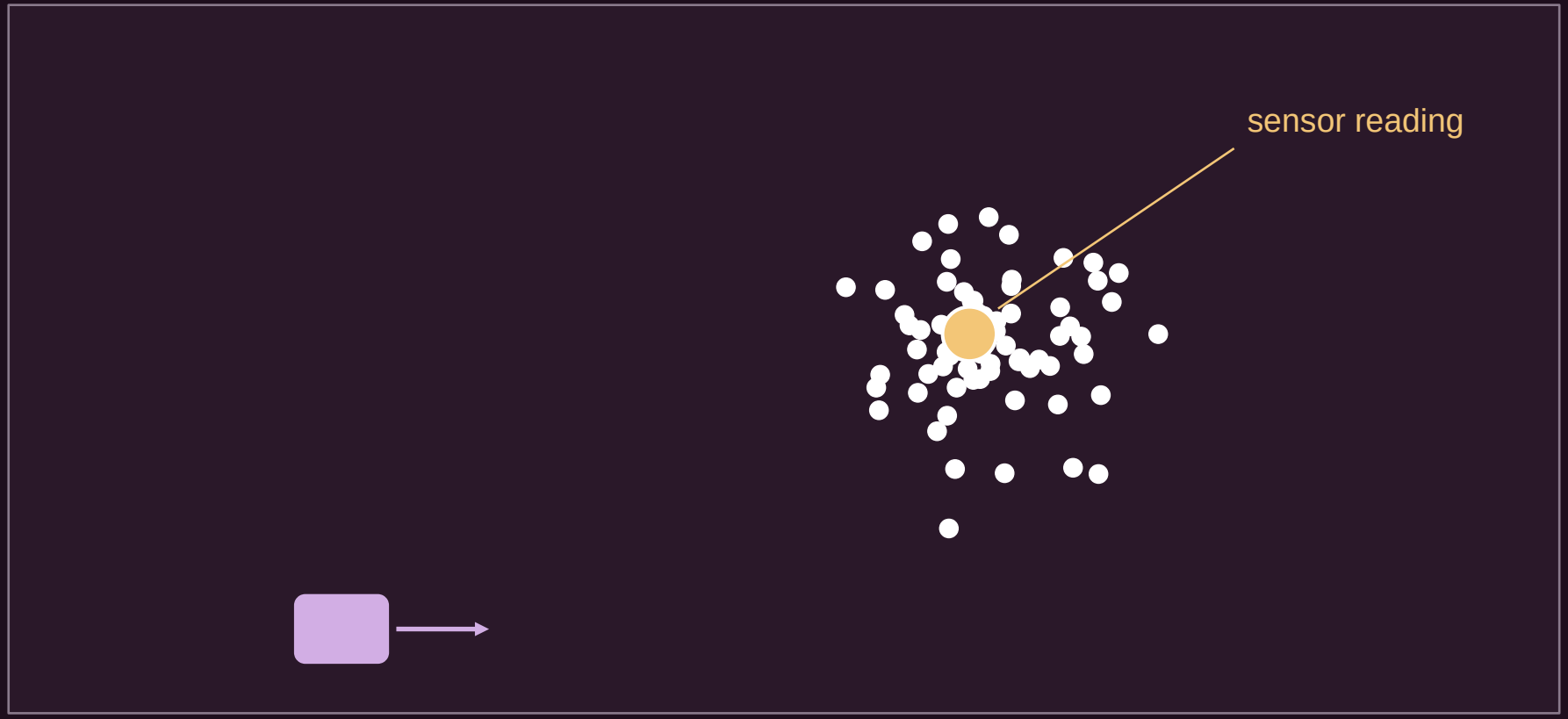
Every particle follows the motion model

Particles: Weight



Particles near the reading are more likely

Particles: Resample



Keep likely particles (with replacement), drop unlikely ones

Particle Filter Algorithm

Algorithm 1: Particle Filter

Given: $f : \mathbb{R}^n \to \mathbb{R}^n$, N motion function, number of particles;
Input: $\mathcal{X}, \mathbf{z}_t, \mathbf{u}_t$: Set of initial particles, measurements, controls

for $x_i \in \mathcal{X}$ **do**
    $x_i = f(\mathbf{x}_i, \mathbf{u}_i) + \mathbf{w}_i$;
    $w_i = P(\mathbf{z}_t | \mathbf{x}_i)$;

$\mathcal{X}_{new} = \emptyset$;
for $m = 1$ *to N* **do**
    $p =$ sample from $\mathcal{X}$ with probability $\propto w_i$;
    add p to $\mathcal{X}_{new}$;

return $\mathcal{X}_{new}$

Summary

Easy to understand and implement

Run time scales with the number of particles

Free choice of motion and sensor models (unlike the Kalman filter)